Test Case 2b: Wing/Body Polar

- **Verification of steady CFD analysis, required**
- **Settings**
  - Steady CFD RANS **French Vanilla SA-[neg] (All terms!)**
    - Adiabatic Wall (not isothermal)
  - Converge residuals to machine precision (~1e-10)
- **Grids: *https://dpw.larc.nasa.gov/DPW8/Scatter/Test_Case_2***
  - NASA CRM geometry including deformed wing matching condition
    - Grid density representative of standard analysis (i.e. L3:**M**edium)

Comparison Data

NTF197: r44,r51,r53
NTF215: r43,r103
NTF229: r296,r300,r302
Ames216: r35,r126,r130,r133

- **Reference Units**

Sref (semi-span grid)	Cref	Semispan	Moment Center
297360.0 sq.in	275.8 in	1156.75 in	(1325.90, 0.00, 177.95)

- **Conditions**

Mach	Re_c	T_{static} (100° F)	γ	Pr	Pr_t	Farfield $\chi = \tilde{\nu}/\nu$
0.85	5×10^6	559.67 R | 310.928 K	1.4	0.72	0.90	3
Polar Conditions		α = [2.50°, 2.75°, 3.00°, 3.25°, 3.50°, 3.75°, 4.00°, 4.25°]				

- **Sutherland's Law**

$$\mu(T) = \mu_0 \left(\frac{T}{T_0}\right)^{3/2} \left(\frac{T_0 + S}{T + S}\right) \quad \mu_0 = 1.716 \times 10^{-5} \frac{\text{kg}}{m\,s} \quad T_0 = 491.6°\,R \quad S = 198.6°\,R \quad \frac{\mu(T)}{\mu_{ref}} = \left(\frac{T}{T_{ref}}\right)^{3/2} \left(\frac{1 + S/T_{fef}}{T/T_{fef} + S/T_{fef}}\right)$$